

# Cloud Computing

## What is Cloud Computing?

Cloud computing is a model that allows network users to access a shared pool of configurable computing resources (like servers, storage, databases, and software) on demand. You can think of it as treating IT infrastructure like a utility—you tap into the resources you need and **pay only for what you use**.

The main **goal** of cloud computing is to allow users to benefit from advanced technologies without needing deep knowledge or expertise in each, which ultimately reduces costs and helps businesses focus on their core mission.

## Cloud - Essentials

Cloud computing relies on two fundamental concepts:

1. **Abstraction:** The system hides the complicated details of the underlying implementation (like where the data is stored or which physical system runs the application) from the users and developers.
2. **Virtualization:** Resources are pooled and shared. Systems and storage can be provisioned (set up) quickly as needed from a centralized infrastructure. The latest NIST definition requires cloud networks to use virtualization and support **multi-tenancy** (sharing resources among two or more clients).

## Cloud Computing - Characteristics and Limitations

Key characteristics include:

- **On-demand self-service**
- **Scalability and Rapid Elasticity** (resources can be quickly grown or shrunk)
- **Resource Pooling**
- **Measured Service** (you pay based on usage)

However, there are **limitations**:

- It requires a constant and reliable Internet connection.
  - There is a **Risk of Vendor Lock-in**, meaning the customer becomes dependent on a single supplier.
  - Security of stored data and potential for unauthorized access remain key concerns.
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# Service Models (The SPI Model)

Cloud computing has three universally accepted **Service Models**, often called the SPI model: **Software, Platform, and Infrastructure**. As you move up the stack (from IaaS to PaaS to SaaS), the vendor manages more, and the customer manages less.

## 1. Infrastructure as a Service (IaaS)

IaaS is the foundation layer. It provides virtualized computing resources, essentially offering **hardware in the cloud**.

- **Concept:** The vendor owns the physical equipment (servers, storage, network infrastructure), and the developer provisions virtual resources (virtual private servers, virtual storage, etc.) as needed.
- **Customer Responsibility:** The client is responsible for installing and managing the **Operating System**, the **Runtime environment**, the **Middleware**, and the **Applications**.
- **Vendor Responsibility:** The vendor manages the underlying infrastructure.
- **Usage:** Used by network architects and skilled developers.
- **Classic Example:** **Amazon Elastic Compute Cloud (EC2)**.

## 2. Platform as a Service (PaaS)

PaaS delivers a complete platform for **developing, deploying, and managing applications**.

- **Concept:** It provides a **ready-to-use environment** that includes virtual machines, operating systems, development frameworks, and tools. This allows developers to focus entirely on coding the application itself.
- **Customer Responsibility:** The client is responsible for installing and managing only the **application** and its **data**.
- **Vendor Responsibility:** The vendor manages the cloud infrastructure, the operating systems, and the enabling software/tools.
- **Benefits:** Reduces coding time by using built-in, pre-coded components (like security and workflow) and supports dispersed or remote development teams.
- **Challenge:** PaaS is difficult because it **locks the developer into a solution dependent upon the platform vendor** (vendor lock-in).
- **Examples:** **Google App Engine, Force.com** (Salesforce's developer platform).

## 3. Software as a Service (SaaS)

SaaS is the most complete cloud computing service model, delivering the entire software solution to the end-user over the Internet.

- **Concept:** Users connect to and use cloud-based applications via a browser. You purchase the solution on a **pay-as-you-go basis**.

- **Customer Responsibility:** The customer's responsibility begins and ends with entering and managing their **data** and user interaction.
- **Vendor Responsibility:** The vendor handles everything: the infrastructure, the software, technical maintenance, security, and updates.
- **Usage:** Popular among consumers and companies; used by the end user.
- **Examples:** **Gmail**, **Microsoft Office 365**, **Salesforce.com** (CRM), and **QuickBooks Online**.

#### Specialized SaaS Topics:

- **Open SaaS:** This is SaaS built on **open source software**, which makes systems cheaper to deploy and reduces vendor lock-in.
- **Mashups:** Componentized SaaS solutions enable mashups, which are applications that display data and features from two or more sources (e.g., annotating a Google map).

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## Cloud Deployment Models

Deployment models define the purpose of the cloud and the nature of how the cloud is located. The NIST definition identifies four types:

| Model                | Definition (Simplified)                                                                                                  | Characteristics                                                                                                    |
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| <b>Public Cloud</b>  | Available for public use, owned by an organization selling cloud services.                                               | Resources shared among multiple customers; high scalability and flexibility; generally less expensive.             |
| <b>Private Cloud</b> | Operated exclusively for a single organization.                                                                          | Resources shared only within that organization; provides predictability and consistency; typically more expensive. |
| <b>Hybrid Cloud</b>  | A combination of two or more clouds (private, community, or public) that remain unique but are bound together as a unit. | Offers flexibility by allowing organizations to mix service types.                                                 |

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| <b>Community Cloud</b> | Organized to serve a common purpose for organizations that share specific concerns (e.g., mission, security, compliance). | The U.S. Government's Apps.gov initiative is an example of a community cloud deployment. |
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## Other Service Models (XaaS)

Many other service models exist under the general term "XaaS" (" as a Service").

### Identity as a Service (IDaaS)

IDaaS companies supply cloud-based **authentication and identity management** to enterprises.

- **Goal:** To ensure users are who they claim to be and grant them the right access to resources.
- **Functionality:** IDaaS may include authentication services, directory services, single sign-on (SSO), and risk monitoring.
- **Benefit:** Provides cost savings compared to managing identity provisioning on-site, and delivers improved, seamless cybersecurity.
- **Interoperability:** Cloud IDaaS systems rely on industry standards for interoperability, such as **OpenID** (for authenticating end-users via a unique URL) and **SAML** (Security Assertion Markup Language, often used for Web Browser Single Sign-On).

### Compliance as a Service (CaaS)

CaaS is a cloud-based model that helps organizations meet regulatory compliance requirements by outsourcing compliance management to a third party.

- **Complexity:** Compliance is highly complex because cloud computing spans different legal jurisdictions.
  - **Application:** CaaS applications often operate as a trusted third party and must manage security policies, privacy, and geographic awareness to be captured in a Service Level Agreement.
  - **Vertical Clouds:** CaaS often appears within "vertical clouds" that specialize in a single market (e.g., athenahealth for medical compliance or bankserv for banking compliance).
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# Cloud Models and Open Standards

## NIST Model

The U.S. National Institute of Standards and Technology (NIST) provides working definitions that separate cloud computing into the Service Models (SPI) and Deployment Models (Public, Private, Hybrid, Community).

## Cloud Cube Model

Maintained by the Jericho Forum, this model categorizes a cloud network based on four dimensions:

1. **Physical location of the data:** Internal (I) / External (E).
2. **Ownership:** Proprietary (P) / Open (O) (a measure of interoperability and vendor lock-in).
3. **Security boundary:** Perimeterised (Per) / De-perimeterised (D-p).
4. **Sourcing:** Insourced / Outsourced (who provides the service).

## Assessing the Role of Open Standards

**Open standards** are publicly available standards developed via a collaborative process, intended to facilitate **interoperability** and data exchange among different products or services.

- Without standards, the industry creates proprietary systems resulting in vendor lock-in.
- The industry is working to create standards-based clouds using architectural guidelines like **Service-oriented architecture** and the deployment of **open-source software**.
- **OpenStack** (a project initiated by Rackspace and NASA) and **Eucalyptus** (a Linux-based platform for IaaS) are examples of open-source projects aiming to create open cloud platform technologies.

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To solidify your understanding of the three main service models (IaaS, PaaS, SaaS), think of them like preparing a pizza:

- **IaaS (Infrastructure as a Service):** You rent the kitchen (the oven, counter, utilities). You bring your own ingredients, recipe, and tools.
- **PaaS (Platform as a Service):** You rent a kitchen that already comes with the pizza oven, flour, cheese, and a standard set of cooking tools. You only need to bring your specific toppings and cook the pizza.
- **SaaS (Software as a Service):** You simply order a fully cooked, ready-to-eat pizza that is delivered to you. You just eat and manage what you eat.

